

MTH102

Linear Algebra

2025–2026–I SEMESTER

01 September 2025

Chapter 1. Preface

This discusses the content covered in the class, along with some examples, exercises, and problem sets. You are encouraged to go through it. Solving the exercises would greatly benefit in understanding the content of the course. The typos/inaccuracies may be reported to jpsaha@iiserb.ac.in.

Timetable

The classes are scheduled on

- Mondays,
- Wednesdays,
- Fridays

from **11:00am to 11:55am** in L5, LHC. The tutorial is scheduled on

- Thursdays

from **5:30pm to 6:30pm** in L5, LHC.

Office hours

The office hours will be held in **Room no. 108, AB1** on

- Mondays, during **5:00pm to 6:00pm**,
- Tuesdays, during **5:00pm to 6:00pm**,
- Wednesdays, during **5:00pm to 6:00pm**
- Fridays, during **6:00pm to 7:00pm**.

The grading policy is as follows.

Components	Weightage
Quiz	20%
Mid-semester examination	35%
End-semester examination	45%

 Calendar

The academic calendar is available at <https://www.iiserb.ac.in/doaa/schedule>.

The announcements related to the course will be made on the following Google classroom page.

<https://classroom.google.com/u/1/c/ODAwOTg3NDg2Mjg0>

You could also access the notes at the link below (until you have joined the Google classroom).

<https://jpsaha.github.io/home/blog/2025/mth102/>

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Introduction

Chapter 2. Sets



Definition of set

A *set* is a well-defined collection of objects. The objects of a set are called its *elements* or *members*.



Examples

- The set of vowels a, e, i, o, u.
- The set consisting of 2, 4, 6, 8, 10.
- The set consisting of 3, 15, 35, 63, 99.

Here are further examples.



Examples

- The set of positive integers divisible by 3.
- The set of prime numbers less than 100.
- The set of positive integers which can be expressed as the product of two distinct primes.
- The set of positive integers which can be expressed as the product of two or more distinct primes.
- The set of positive integers which are smaller than 100 and share no common factor with 100.



Remark

Note that the first few sets are described by writing down all its elements, whereas in the latter examples, the sets are described by the rules or properties which determine whether a particular object is an element or not.

☰ Notation

A set is usually denoted by an uppercase letter, for example,

$$A, B, C, X, Y, Z,$$

whereas the lowercase letters are used to denote its elements. For instance, one may use

- a to denote an element of A ,
- b to denote an element of B ,
- c to denote an element of C ,
- x to denote an element of X ,
- y to denote an element of Y ,
- z to denote an element of Z .

Several special sets are denoted by certain standard symbols. Here are some such examples.

$\mathbb{N}$	the set of positive integers
$\mathbb{Z}$	the set of integers

If x is an element of a set X , then one says that x **belongs to** X , and writes

$$x \in X.$$

Note that 2 belongs to $\mathbb{N}$, but $\frac{1}{2}$ does not belong to $\mathbb{N}$. One writes

$$2 \in \mathbb{N}, \frac{1}{2} \notin \mathbb{N}.$$

Notation

There are two ways to write down a set. For example, consider the set of vowels, which is written as

$$V = \{a, e, i, o, u\}.$$

Note that the elements are separated by commas and enclosed in braces $\{\}$.

Another way to write down a set is by stating the *rules* or *properties* which determine whether a particular object is an element or not. For example, the set of even positive integers is written as

$$E = \{x \in \mathbb{Z} : x \text{ is even, } x > 0\}.$$

This reads

“ E is the set of elements x in $\mathbb{Z}$ such that x is even and $x > 0$ ”.

Example

Note that the set $A = \{3, 15, 35, 63, 99\}$ is equal to

$$\{x \in \mathbb{Z} : x \text{ is equal to } n^2 - 1 \text{ for some } n \in \{2, 4, 6, 8, 10\}\}.$$

Since 1, 2 are the roots of the polynomial $x^2 - 3x + 2$, it follows that

$$\{x \in \mathbb{N} : x^2 - 3x + 2 = 0\} = \{1, 2\}.$$

Also note that

$$\mathbb{N} = \{1, 2, 3, \dots\},$$

$$\mathbb{Z} = \{0, 1, -1, 2, -2, 3, -3, \dots\}.$$

One uses the symbol $:=$ to indicate that the symbol on the left is being defined by the symbol on the right.

$$\mathbb{Q} := \left\{ \frac{m}{n} : m, n \in \mathbb{Z} \text{ and } n \neq 0 \right\}.$$

It is called the *set of rational numbers*.

If the number of elements of a set is finite, then it is called a **finite set**. If a set is not finite, it is called an **infinite set**.



Example

The sets

$$\{a, e, i, o, u\}, \{3, 15, 35, 63, 99\}, \{x \in \mathbb{N} : x^2 - 3x + 2 = 0\}$$

are finite. The sets $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ are infinite.

Exercise 2.1. Determine the elements of the following sets.

- $\{x \in \mathbb{N} : x^2 - 1 = 0\}$.
- $\{x \in \mathbb{Z} : x^2 - 1 = 0\}$.

§2.1 Basic terminologies

If A, B are sets, and every element of A also belongs to B , then we say that A is a **subset** of B , and write

$$A \subseteq B.$$

If A is not a subset of B , one writes

$$A \not\subseteq B.$$



Example

- Note that every set is a subset of itself.
- Note that $\mathbb{N}$ is a subset of $\mathbb{Z}$, and $\mathbb{Z}$ is a subset of $\mathbb{Q}$. One writes

$$\mathbb{N} \subseteq \mathbb{Z}, \quad \mathbb{Z} \subseteq \mathbb{Q}.$$

- Also note that

$$\mathbb{Z} \not\subseteq \mathbb{N}, \mathbb{Q} \not\subseteq \mathbb{Z}, \mathbb{Q} \not\subseteq \mathbb{N}.$$